Supplementary Information for
Proteus: Adaptive Scale-Space Analysis for Learning Non-Linear Fuzzy Manifold Memberships

S1. Initial Dimensionality Reduction

Randomized SVD: Sample up to 20 000 rows (or all if small), compute a randomized SVD with sketch width $k_{\text{target}} \approx 256$ and oversampling 20. Determine rank by cumulative energy $\geq 99.9\%$ and/or Levina–Bickel intrinsic dimension; keep $r = \max(r_{0.999}, 2\lceil D_{\text{LB}} \rceil)$ if it yields a $\geq 5\%$ reduction; otherwise skip projection. Optionally append 32 JL vectors to tighten distance preservation.

$$P = V_{1:r}^{\top} \;\Rightarrow\; Z = X\,P\,, \quad Z \in \mathbb{R}^{N \times r}.$$

Rationale: Cuts ANN costs and per-sample updates by $\times 5$–$\times 40$ on high-d data while retaining $\geq 99.9\%$ variance and local distances.

S1.1 Optional Pre-processing via Marginal Gaussianisation

Workflow: For datasets where individual features exhibit high skew or heavy tails, an optional pre-processing step can significantly accelerate convergence. For each feature x_j, its empirical CDF $\hat{F}_j$ is computed on a data subsample. The features are then transformed via the inverse error function to be marginally Gaussian: $z_j = \sqrt{2}\,\text{erf}^{-1}(2\hat{F}_j(x_j) - 1)$.

Linearity Test and Model Selection: After this transform, statistical linearity tests (e.g., Graph-Laplacian Curvature Energy and distance correlation) are run on the z-space. If the data is found to be linearly separable to within a given tolerance ($p \geq 0.01$), the expensive fitting of a global `Glow` model can be skipped entirely for that recursion level, and the algorithm can proceed directly to the geometric audit. This provides a principled mechanism for avoiding unnecessary non-linear transformations.

S2. First-Principles Derivations

S2.1 Geometric Grid Spacing

Normalized responses in scale-space obey $\Delta_{\text{FWHM}}(\log\sigma) \approx 0.586$. If two peaks are separated by less than this, they are not reliably distinguishable. Therefore, choose adjacent scales to differ by $\log\sigma_{j+1} - \log\sigma_j = \Delta \approx \frac{1}{2}\Delta_{\text{FWHM}}$, giving $\sigma_{j+1}/\sigma_j = e^{\Delta} \approx \sqrt{2}$. Since $\tau = \sigma^2$, the geometric ratio in τ becomes $r = \tau_{j+1}/\tau_j = (\sigma_{j+1}/\sigma_j)^2 \approx 1/2$. In practice we span both directions, so we use $r = 1/\sqrt{2} \approx 0.71$ in σ (7–9 points/decade) and map consistently into τ.

S2.2 EWMA Weight from Neighborhood Size

EWMA recursion $m_{t+1} = (1-\alpha)m_t + \alpha x_{t+1}$ has half-life $T_{1/2} = \ln(0.5)/\ln(1-\alpha) \approx \ln 2/\alpha$ for small α. Set $T_{1/2}=k$ (one neighborhood of updates) $\Rightarrow \alpha = \ln 2/k$. This ties estimator memory to the actual neighborhood support.

S2.3 Dual-Rate Motion from Variance Correction

Under isotropy, variance $\sigma_i^2 \approx \mathbb{E}\|x - w_i\|^2$. A radial shift δr changes variance by $\Delta\sigma_i^2 \approx 2\sqrt{\sigma_i^2}\,\delta r$. Spreading over $H{=}k$ effective updates gives

$$\eta_{\mathrm{GNG},i} \approx \frac{\Delta\sigma_i^2}{2H\,\sigma_i^2} = \frac{1}{2k}\Big(1 - \frac{\sigma_i^2}{\tau}\Big); \qquad \eta_{\mathrm{GNG},i} = \frac{\ln 2}{2k}\Big(1 - \frac{\sigma_i^2}{\tau}\Big) \text{ with half-life normalization.}$$

For centering, with $\delta_{\min} = \kappa(1{-}r)\sqrt{\tau}$ and expected drift per update $\approx \sqrt{\tau}$, set $k\,\eta_{\mathrm{cent}}\,\sqrt{\tau} = \delta_{\min}$ $\Rightarrow \eta_{\mathrm{cent}} = \kappa(1{-}r)/k$. Unless stated otherwise, we use a default of $\kappa = 0.5$.

Deferred nudge mechanism: Node positions are updated only when the accumulated nudge $\mathbf{a}_i$ exceeds the resolution threshold $\delta_{\min}$, ensuring mesh changes occur only when meaningfully resolvable at the current scale. This prevents high-frequency noise from destabilizing the mesh while maintaining geometric accuracy.

S2.4 Control-to-Threshold Mapping

Constrain growth to local intrinsic dimension by mapping optimizer control to thresholds: $\tau_{\mathrm{global}} = -D_{\mathrm{subspace}}\,\log(1{-}s_{\mathrm{control}})$ and $\tau_{\mathrm{local},i} = -d_{\mathrm{final},i}\,\log(1{-}s_{\mathrm{control}})$.

S2.5 Scale Response $\Phi(\tau)$: Canonical Formula

Baseline: uniform-density scaling. At equilibrium, a variance-cap GNG quantizes a probability density into cells each containing variance $\approx \tau$. For a uniform density on a bounded D-dimensional domain of volume Vol, each stabilized cell has linear size $\sim \sqrt{\tau}$ and volume $\sim \tau^{D/2}$, so the equilibrium node count is

$$N_{\mathrm{uniform}}(\tau) \;\approx\; c_D\,\mathrm{Vol}\,\tau^{-D/2}\,, \qquad \frac{d\log N_{\mathrm{uniform}}}{d\log\tau} \;=\; -\frac{D}{2}\,.$$

This gives a sharp *null hypothesis*: featureless data produces a mesh whose size scales as $\tau^{-D/2}$ on a log-log plot, with slope exactly $-D/2$. Any deviation from this slope is evidence of data structure at that scale.

Canonical $\Phi(\tau)$. Let the controller evaluate a geometric grid $\{\tau_j\}$ (ratio $r = 1/\sqrt{2}$, S2.1). At each τ_j, run Stage 1 to stabilization yielding $N(\tau_j)$ equilibrated nodes and a coefficient-of-variation diagnostic $\mathrm{CV}(h;\tau_j)$. Define the centered log-log slope

$$s(\tau_j) \;=\; \frac{\log N(\tau_{j+1}) - \log N(\tau_{j-1})}{\log\tau_{j+1} - \log\tau_{j-1}}\,,$$

using one-sided differences at the grid endpoints. The scale response is

$$\boxed{\Phi(\tau_j) \;=\; \max\!\big(0,\; s(\tau_j) + D_{\mathrm{subspace}}/2\big)\cdot\mathbf{1}\big[\mathrm{CV}(h;\tau_j) < \varepsilon_{\mathrm{CV}}\big]\cdot\mathbf{1}\big[N(\tau_j) \geq N_{\min}\big]}$$

with $\varepsilon_{\mathrm{CV}} = 0.01$ (S10) and a regularity floor $N_{\min}$ (default 10). The first indicator gates on Stage 1 having actually converged; the second rejects the trivial regime where the mesh has collapsed to one or two nodes.

Interpretation.

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- **Uniform data:** $s(\tau) = -D/2$ at all scales $\Rightarrow \Phi \equiv 0$.

- **Single characteristic scale σ_*:** as τ decreases through σ_*^2, the slope $s(\tau)$ flattens from $-D/2$ toward 0 (a "plateau" in N as clusters at scale σ_* stabilize into roughly one node each), then steepens back toward $-D/2$ when $\tau < \sigma_*^2$ starts resolving within-cluster noise. Φ exhibits a positive peak centered at $\tau \approx \sigma_*^2$, with magnitude bounded by $D/2$.
- **Multi-scale data:** each characteristic scale contributes a distinct peak; the FWHM$(\log \sigma) \approx 0.586$ resolution argument of S2.1 guarantees separable peaks.

Properties.

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- *Dimensionless and scale-comparable.* Φ is a slope anomaly, invariant under rescaling of τ.
- *Bounded.* $\Phi \in [0, D_{\text{subspace}}/2]$ by construction.
- *Zero on featureless data.* Eliminates spurious peaks from uniform regimes.
- *Cheap.* One scalar per grid point; the dominant cost is the Stage 1 run itself, not the response computation.

Bracketing and refinement. The controller identifies candidate characteristic scales as local maxima of Φ on the grid. A candidate at τ_j is retained if $\Phi(\tau_j) > \Phi_{\min}$ (default $0.1 \cdot D/2$) and exceeds both neighbors. Brackets are formed from adjacent retained maxima and refined by a lightweight Bayesian optimizer (e.g., Gaussian-process surrogate with five extra τ evaluations per bracket) restricted to the bracketed interval.

Relationship to Lindeberg scale-space. Φ is a discrete, mesh-level analogue of Lindeberg's scale-normalized derivatives with $\gamma = 2$, the optimal choice for detecting Gaussian-kernel features (Lindeberg 1998). Our setting inherits this because the variance-cap GNG quantizes variance, not an arbitrary functional: at equilibrium $\sigma_i^2 \approx \tau$ in each cell, and the only remaining scale-dependent statistic (mesh size $N(\tau)$) has a closed-form $\tau^{-D/2}$ baseline.

Degenerate cases and fallbacks.

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- *Stage 1 fails to stabilize.* CV $\geq \varepsilon_{\text{CV}}$ forces $\Phi(\tau_j) = 0$. If stabilization fails across a contiguous stretch of the grid, the controller escalates: double the stabilization budget (S10) and retry; if still failing, declare the region terminal.
- D_{subspace} *poorly estimated.* At the first recursion level, D_{subspace} is taken from the initial Levina–Bickel estimate (S1). An error of ± 1 shifts the baseline slope by ± 0.5 and can generate spurious small peaks; the $\Phi_{\min}$ threshold absorbs this.
- *Very small datasets.* When N_{samples} is small, $N(\tau)$ saturates at a few nodes over a wide τ range, making $s(\tau) \approx 0$ artifactually. Set $N_{\min} = \max(10, \sqrt{N_{\text{samples}}})$ in that regime.

S3. Statistical Gauntlets

Unless noted, we adopt significance defaults consistent with Stage 1: Welch's t tests at p=0.05 (one-tailed where directional) for neighborhood-stability triggers, and Wilson one-sided bounds at 90% (i.e., z=1.2816) for link significance comparisons.

S3.1 Link Pruning

Power shield: protect a new link until $n_{\text{eff}} \geq n_{\text{req}}$ from an analytic power formula for proportions. *Relative weakness:* compute Wilson one-sided upper bound ν^+ for link proportion p with count n:

$$\nu^+ = \frac{p + z^2/(2n) + z\sqrt{p(1-p)/n + z^2/(4n^2)}}{1 + z^2/n}, \ z{=}1.2816.$$

The Wilson score interval is used in place of a standard t-test as it is an "exact" test for binomial proportions that does not assume normality, making it robust and accurate even for the low-hit-count links common in early training. Vote to prune if $\nu^+ <$ median peer significance; require bilateral agreement from both endpoints.

S3.2 Node Pruning

Trigger: require neighborhood stability (Welch t on $\tilde{\rho}$). *Fair chance:* Poisson arrival-rate test $\lambda_{\text{est}} = h_i/\text{lifetime} \geq 2/\delta_{\min}$. *Relative:* prune if hits$<$ 0.5$\times$ neighbor mean *and* variance $\sigma_i^2 < 0.5\,\tau_{\text{local},i}$.

The composite rule for node pruning (requiring both low relative hits *and* low local variance) is a critical safeguard. It ensures the system only removes nodes that are both statistically insignificant (low-mass) and located in regions that have already been sufficiently linearized. This prevents the algorithm from mistakenly punching holes in the manifold in areas that are still curved and under active refinement.

S3.3 Node Pruning (Stage 2, simplex-arbitrated)

Arbitration by simplex star: A deletion proposal for node i is decided by a vote over its incident simplexes $\mathcal{S}(i)$; require majority agreement based on local stability and activity metrics. *Geometric veto:* Always perform a reconstruction error test; if removal of i causes a significant increase in local reconstruction error (tent-pole behavior), veto the deletion. *Rationale:* Stage 2's geometry is mediated by the simplicial complex; arbitration by $\mathcal{S}(i)$ preserves structural integrity in complex regions.

S3.4 Evidence-Based Structure Learning (Local Evidence Score)

Purely geometric triggers (e.g., $\sigma_i^2 > \tau_{\text{local},i}$ or high torsion ratio R_S) are fast indicators of local misfit, but they cannot distinguish unresolved structure from noise under weak evidence. Proteus therefore treats geometry as a *proposal* mechanism and uses a local evidence test to *ratify* discrete edits (splits, pruning/merges, and optional warp attachments). The evidence is computed from how well the current topology explains the observed local transition counts $\{n_{i\to j}\}$ under the geometry-induced router $q(j \mid i; m)$ (see S13). The score $F(\mathcal{R}; M)$ defined below is a *leading-order Laplace approximation of the log marginal likelihood* restricted to the affected region; the "free energy" name is retained as a conventional label but refers here to Bayesian model-selection evidence, not to variational free energy (ELBO).

Affected region. Let M denote a candidate topology (keep vs. edit). For a proposed edit, define the affected node set $\mathcal{V}_{\text{aff}}$ as the set of nodes whose simplex stars change under the edit, augmented by a small neighbor ring to account for router denominators:

$$\mathcal{V}_{\text{aff}} \;=\; \mathcal{V}_{\text{core}} \,\cup\, \bigcup_{i\in\mathcal{V}_{\text{core}}} \mathcal{N}(i)\,, \qquad \mathcal{V}_{\text{core}} \;=\; \{i : \text{Star}_M(i) \text{ differs between candidates}\}\,.$$

Let the affected simplex set be $\mathcal{S}_{\text{aff}} = \bigcup_{i\in\mathcal{V}_{\text{aff}}} \text{Star}_M(i)$. We treat the exterior mass field as fixed boundary conditions: m_S for $S \notin \mathcal{S}_{\text{aff}}$ is held at its current settled value.

Local likelihood and prior. Restrict the transition log-likelihood to affected sources:

$$\mathcal{L}_{\mathcal{R}}(m;M) \;=\; \sum_{i\in\mathcal{V}_{\text{aff}}} \;\sum_{j\in\mathcal{N}(i)} n_{i\to j}\, \log q(j \mid i; m, M)\,,$$

where q is computed using the full denominator for each i but with exterior masses frozen as above. Choose a prior $\psi(m \mid M)$ (e.g., Dirichlet composition on simplex masses, or logistic-GMRF smoothness on logits as in S13.3).

Local settle (MAP). For each candidate topology M, warm-start the affected masses from the current solution and settle to a local MAP configuration:

$$\hat{m}_M \;=\; \arg\max_{m\in\Delta}\; \mathcal{L}_{\mathcal{R}}(m;M) \;+\; \log\psi(m \mid M)\,,$$

holding exterior masses fixed. In practice, settling is performed with a fixed small iteration budget for comparability across candidates (and because the acceptance test only requires a reliable local ranking).

Local evidence score (Laplace/BIC form) and acceptance. We score each candidate by the negative log joint at the MAP, plus a BIC-style complexity correction:

$$F(\mathcal{R};M) \;=\; -\mathcal{L}_{\mathcal{R}}(\hat{m}_M;M) \;-\; \log\psi(\hat{m}_M \mid M) \;+\; \tfrac{1}{2}\,k_M\,\log N_{\text{eff},\mathcal{R}}\,,$$

where $N_{\text{eff},\mathcal{R}} = \sum_{i\in\mathcal{V}_{\text{aff}}}\sum_j n_{i\to j}$ and k_M is the number of free mass parameters in the affected region (typically $k_M = |\mathcal{S}_{\text{aff}}| - 1$ under the simplex constraint, counted consistently across candidates). The final term is the asymptotic (large-N_{eff}) limit of the Laplace approximation of the log marginal likelihood; equivalently, $F(\mathcal{R};M) \approx -\log p(\text{data} \mid M, \mathcal{R})$ up to an additive constant. The edit is accepted iff F decreases by a margin:

$$F(\mathcal{R};M_{\text{edit}}) \;<\; F(\mathcal{R};M_{\text{keep}}) \;-\; \log\tau\,,$$

with $\tau > 1$ controlling conservatism (e.g., $\log\tau \in [1,3]$ for splits; merges/prunes may use $\tau = 1$). Decreasing F is equivalent to increasing approximate log evidence; the log-ratio threshold $\log\tau$ is a log-Bayes-factor margin.

When BIC is loose: substitutes. The BIC form assumes standard regularity (posterior concentration, interior MAP, nonsingular Hessian). Two principled refinements are available: *(i) Full Laplace.* Replace $\frac{1}{2}k_M\log N_{\text{eff},\mathcal{R}}$ with $\frac{1}{2}\log\det(H_M/(2\pi))$, where $H_M = -\nabla_m^2[\mathcal{L}_{\mathcal{R}} + \log\psi]\big|_{\hat{m}_M}$. This corrects the leading constant in the Laplace approximation and is more accurate when $\hat{m}$ is near a simplex boundary or when the effective sample size is modest. *(ii) Dirichlet–multinomial (DM) marginals.* If the prior is placed directly on the per-node transition distribution $q(\cdot \mid i) \sim \text{Dir}(\alpha_0\mathbf{1})$ rather than on m (cf. S13), the marginal likelihood at each node is analytic:

$$p(\{n_{i\to j}\} \mid M, \alpha_0) \;=\; \frac{\Gamma(J\alpha_0)}{\Gamma(J\alpha_0 + n_i)} \prod_{j=1}^{J} \frac{\Gamma(\alpha_0 + n_{i\to j})}{\Gamma(\alpha_0)}\,,$$

and the per-region score becomes $F_{\text{DM}}(\mathcal{R};M) = -\sum_{i\in\mathcal{V}_{\text{aff}}} \log p(\{n_{i\to j}\} \mid M, \alpha_0)$. This eliminates the regularity caveat (no Laplace approximation needed) and makes the Occam factor an intrinsic consequence of the Dirichlet prior volume. *(iii) Predictive checks.* WAIC and PSIS-LOO provide finite-sample predictive-accuracy alternatives and correspond to Bayes-factor proxies via BF $\approx \exp(\Delta\text{ELPD})$. We use BIC as the default (cheapest); full Laplace or DM substitutes are used when BIC's assumptions are in doubt.

S4. Torsion and Shape Quality

The geometric audit operates on the principle that error responses depend on local rigidity: in *plastic* regions, error produces geometric adjustments; in *rigid* regions, error accumulates as structural stress requiring topological resolution. Let $\bar{\mathbf{w}}$ and $\bar{\mathbf{m}}$ be barycenters. With $E = [\mathbf{w}_{v_1}-\bar{\mathbf{w}} \ \cdots \ \mathbf{w}_{v_d}-\bar{\mathbf{w}}]$ and $M = [\mathbf{m}_{v_1}-\bar{\mathbf{m}} \ \cdots \ \mathbf{m}_{v_d}-\bar{\mathbf{m}}]$, define

$$\Omega_S = M^\top E - E^\top M\,, \qquad \kappa_S = \|\Omega_S\|_F\,; \qquad R_S = \kappa_S/\tau^*.$$

Torsion Ladder: *Ignore* if $R_S<0.05$ (numerical noise). *Keep as-is* if $0.05 \leq R_S<0.30$ (tolerable curvature). *Geometric fix* if $0.30 \leq R_S<0.60$: propose a torsion-aligned split with explicit placement below, and accept it only if it passes the local evidence test in S3.4 (in addition to the numerical safeguards in this section). *Non-linear upgrade* if $R_S \geq 0.60$: attach a local mini-NSF (see S7) to the affected patch. If the fitted mini-NSF fails to reduce R_S below 0.60 on the patch, propose a torsion-aligned split as a geometric fallback, again gated by S3.4.

A split is only permitted if the parent simplex (or node) has accumulated sufficient hits to pass a budget check ($h_{\text{parent}} \geq 2{\cdot}h_{\text{prune}}$), ensuring that the resulting children are not immediately vulnerable to pruning under the framework's statistical gauntlets.

Split placement: Insert the new node at the midpoint of the simplex's longest edge projected onto the principal axis of Ω_S (dominant torsion axis). If any child has $Q_S<0.25$, redo via centroid star split.

Shape quality safeguard: The simplex shape metric $Q_S = d\,r_{\text{in}}/R_{\text{circ}} \in (0,1]$ uses the inradius r_{in} and circumradius R_{circ}. Poor quality ($Q_S < 0.25$) indicates numerical instability risk; the centroid split fallback ensures stable face normals and well-conditioned gradients.

S5. Dual Flow Mechanisms

Accumulate facet pressures for winning simplex S via $\Delta p_f \propto \max(0, (\mathbf{x}-\bar{\mathbf{w}}_S)^\top \mathbf{n}_f)$. This *fractional tallying* avoids winner–take–all artifacts and yields smoother, more stable boundary pressure estimates than single-face voting. On the dual graph, solve

$$\min_{\{p_f\}} \ \lambda \sum_f (p_f-\hat{p}_f)^2 + \mu \sum_S \|A_S\,\mathbf{p}_S\|_2^2\,,$$

with Gauss–Seidel iterations $p_f \leftarrow (\lambda\hat{p}_f + \mu \sum_{S\ni f}(A_S^\top A_S\,\mathbf{p}_S)_f)/(\lambda + \mu\ \deg(f))$, or equivalently by loopy BP messages until convergence. This reconstruction is conditioned on the accepted topology and its settled geometry/mass field; it does not act as an additional likelihood term for the same simplex masses.

Definition of A_S (discrete divergence stencil). Let simplex S have $d{+}1$ facets $f_0, \ldots, f_d$ with outward unit normals $\mathbf{n}_{f_i} \in \mathbb{R}^d$ and facet (hyper)areas A_{f_i}. Stack the pressures incident to S as $\mathbf{p}_S = (p_{f_0}, \ldots, p_{f_d})^\top \in \mathbb{R}^{d+1}$, and define $A_S \in \mathbb{R}^{d\times(d+1)}$ column-wise by

$$(A_S)_{\cdot,i} \ = \ A_{f_i}\,\mathbf{n}_{f_i}\,, \qquad i = 0, \ldots, d\,.$$

Then $A_S\,\mathbf{p}_S = \sum_i A_{f_i}\,p_{f_i}\,\mathbf{n}_{f_i}$ is the discrete-divergence-theorem sum over the facets of S: a vector in $\mathbb{R}^d$ measuring the net oriented flux leaving S under the pressure configuration. At steady state on a conservative field this sum vanishes, so the penalty $\|A_S\,\mathbf{p}_S\|_2^2$ enforces discrete conservation cell-by-cell. Equivalently, A_S is the matrix of the discrete (primal) divergence operator restricted to S; $A_S^\top A_S \in \mathbb{R}^{(d+1)\times(d+1)}$ is the local stencil used in the Gauss–Seidel update above. For boundary faces (facets with no neighboring simplex on one side), the corresponding column is zeroed to impose the Neumann-like no-exterior-flux convention (see S9).

S6. Masking for Missing Modalities

For input $\mathbf{x} \in \mathbb{R}^d$ with structured sparsity, build boolean mask $\mathbf{m} = (x \neq 0)$. Compute distances and residuals only on active dimensions: $\|\mathbf{x}-\mathbf{w}\|^2 = \sum_{j:m_j}(x_j-w_j)^2$; residual $\mathbf{e} = \mathbf{x}-\mathbf{w}$ with e_j=0 if m_j=0. Apply the mask to all vector updates (moments, nudges, Oja), and accumulate variance only on active dimensions. Optionally normalize updates by the active count to equalize per-sample influence. This preserves geometry and statistics in multi-modal fusion where zeros indicate missingness rather than small values.

S7. Non-Linear Warp Strategies

When geometric refinement (node moves and splits) is insufficient to resolve high residual curvature, Proteus can apply a non-linear warp. The core strategy is a hybrid approach that adapts to the nature of the curvature, measured by the torsion coverage P_κ (the fraction of simplices with torsion ratio $R_S \geq 0.30$).

Global vs. Patch-wise Strategy:

- If curvature is **widespread** ($P_\kappa > 50\%$), a shallow *global Glow* is trained. Its invertible 1×1 convolutions mix dimensions efficiently, providing a good baseline correction for system-wide bends.
- If curvature is **patchy** ($P_\kappa \leq 25\%$), compact and expressive *mini-Normalizing Spline Flows (NSFs)* are attached only to the highest-torsion patches. This avoids wasting model capacity on regions that are already linear.
- If curvature is **mixed** ($25\% < P_\kappa \leq 50\%$), a moderate global Glow is combined with mini-NSFs on the worst ∼10% of patches.

Patch Identification and Mini-NSF Training: A facet-adjacency graph is built over all "hot" simplices ($R_S \geq 0.60$). Leiden community detection finds contiguous patches of high torsion, with resolution auto-tuned to ensure no patch exceeds a size cap $P_{\max}$ (e.g., 64 simplices). Each mini-NSF is a compact model (2-3 layers of rational-quadratic spline couplings, 64-128 hidden units, 8-12 bins) trained for a small number of epochs on its routed data, with the global model frozen. If, after training, any simplex in the patch still exhibits $R_S \geq 0.60$, attempt a torsion-aligned split as a geometric fallback.

S8. Numerical Stability and Implementation

Warm-start and state reset: On entering Stage 2 at fixed τ^*, shrink EWMA moments by a decay factor $\gamma_0 \approx 0.7$, carry hit/link counts unchanged, and reset the nudge accumulator $\mathbf{a}$ to zero to ensure stable dynamics under the richer neighborhood model ($k \approx d$). **Conditioning for Ω_S:** Use QR (preferred) or thin SVD to stabilize E; scale residuals M by a local norm to avoid magnitude blow-up; compute $\Omega_S = M^T E - E^T M$ from stabilized factors; cache per-simplex factors and recompute only after geometric moves/splits.

Small constants and clipping: Set ε in ρ_i to $10^{-8}-10^{-6}$ depending on data scale; clip η_{GNG} to $|\eta| \leq 0.3$ (Stage 1) and ≤ 0.05 (Stage 2); constrain spline slopes/bins to avoid near-singular Jacobians; bound $\|\mathbf{a}_i\|$ growth between apply steps.

Cache face normals: Precompute and cache per–simplex face normals; recompute only after deferred moves or splits affecting a vertex. This keeps torsion and face-pressure updates $\mathcal{O}(d^2)$ and stable.

ANN configuration (HNSW): Typical defaults: M=16 − 32, efConstruction=100 − 400, efSearch=50 − 200; increase efSearch modestly near convergence to improve neighbor stability. Persist indices per recursion level; rebuild only after significant mesh mutation.

S8.1 Practical Implementation Notes

Rank-1 Oja vs. Scalar Variance. While the core theory holds with a simple scalar variance tracker, the specified implementation uses a rank-1 Oja's rule update to track the dominant principal component of the local error. This provides a more accurate, data-driven vector for placing new nodes during a split, at the cost of one extra stored vector per node. For ambient dimensions $d \lesssim 10,000$, the improved split quality justifies the modest increase in memory, and this method is recommended.

Initial Simplex Seeding. Upon entering Stage 2, the initial simplicial complex is seeded from the Stage 1 node-edge graph using a fast, "Greedy Chaining" heuristic. This heuristic rapidly discovers the majority of d-simplices, providing a robust starting point for the dynamic refinement process.

S8.2 Cost Envelopes

Stage 1 per-epoch time is $\mathcal{O}\big(N\,d\,k\,\log N_{\text{nodes}}\big)$ with HNSW queries; space is $\mathcal{O}(N_{\text{nodes}}\,d + N_{\text{nodes}}\,\bar{d})$. Stage 2 initial simplex seeding by Greedy Chaining is $\mathcal{O}\big(N_c\,d_{\text{cand}}^2\,d_c\big)$; torsion audits are $\mathcal{O}\big(N_{\text{simplices}}\,(d{+}1)^2\,d\big)$. The dual graph used by the belief propagation solver has size $\mathcal{O}(N_{\text{simplices}}\,d)$.

S9. Robustness and Edge Cases

Outliers: Use Huber-like weighting on residuals in moment updates or winsorize top 0.5% residual norms; optionally cap per-sample hit contributions.

Boundaries and orientation: For open meshes, treat boundary faces with Neumann-like conditions in the dual flow (no flux across exterior). In non-orientable regions, operate on local orientation patches to keep face normals consistent.

Sparse patches: Require a minimum routed sample count $n_{\min} = \max(10\,d, 1000)$ before fitting a mini-NSF; otherwise defer to geometric refinement.

S10. Evaluation Protocols

Defaults: Stage 1 k=8, α= ln 2/8, grid r=1/$\sqrt{2}$; Stage 2 $k{\approx}d$, α= ln 2/k. Stopping: CV< 0.01 and R_S<0.30. PH: VR complex, max dim 2, filtration up to $1.5\,\sigma_\star$. MMD: RBF kernel, bandwidth median heuristic, 1 000 samples/1 000 tests. Ablations: remove Stage 1 scaffold, Torsion Ladder, or Dual Flow.

Persistent Homology: Vietoris–Rips (max dim 2), filtration to $1.5\,\sigma_\star$; report bottleneck/2-Wasserstein. **MMD:** RBF kernel (median heuristic), 1000 samples/1000 tests, 5 seeds. **Log-likelihood:** Sum of face-pressure potentials within simplex plus normalization; use held-out data.

S11. Algorithmic Details

The following procedures reference node and link bookkeeping commonly used in the implementation: nodes track an *update count* and creation index N_{creation} in addition to positions and EWMA statistics; links maintain directed counters and recent activity snapshots. These fields support pruning power analyses and tenure checks but are not essential for the mathematical description.

Flow-conserving initialization after a split. Let C be the parent simplex with vertices $V(C) = \{v_1, \ldots, v_{d+1}\}$ and let $h_{\text{old}}(v_i, v_j)$ denote pre-split internal hit counts between vertices. When inserting a new vertex k, initialize the new tallies by

$$h_{\text{new}}(k, v_i) \;=\; \tfrac{1}{2} \sum_{\substack{v_j \in V(C) \\ j \neq i}} h_{\text{old}}(v_i, v_j)\,.$$

This preserves the total internal mass $H_{\text{total}}(C)$ and approximately preserves the local flow gradient.

Data structures (Stage 2 additions). Each simplex maintains small fields to support the torsion audit and dual-flow:

- `face_pressures`: array of size $d{+}1$ for facet pressures
- `torsion_sum`, `torsion_sum_sq`: accumulators for κ_S
- `simplex_update_count`: number of updates contributing to the audit

Algorithm 1 Stage 1 at Fixed τ

procedure RunToEquilibrium(X, τ)
  Initialize mesh $\mathcal{M} \leftarrow \varnothing$ ▷ Create links between BMU_1 and BMU_2 on first co-activation; maintain directed weighted counters C_{12}, C_{21}
  repeat
    Sample $\mathbf{x}$ from X
    $\mathcal{N}_k \leftarrow$ QueryANN($\mathbf{x}, k$) ▷ Find k nearest nodes
    ApplyRankWeightedUpdates($\mathbf{x}$, $\mathcal{M}$, $\mathcal{N}_k$)
    for each updated node i **do**
      if $\|\mathbf{node}_i.\mathbf{a}\| \geq \delta_{min}$ **then** ▷ Deferred move
        $\mathbf{node}_i.\mathbf{w}$ += $\mathbf{node}_i.\mathbf{a}$; $\mathbf{node}_i.\mathbf{a} \leftarrow 0$; RefreshANN($\mathbf{node}_i$)
      end if
      if $\mathbf{node}_i.\sigma^2 > \tau_{local,i}$ **then** ▷ Split rule
        SplitNode($\mathbf{node}_i$) ▷ Flush nudge; place child at $\mathbf{w}_i{+}\mathbf{m}_i$ along $\mathbf{u}_i$; shrink-inherit stats
      end if
    end for
    ApplyPruningGauntlets($\mathcal{M}$) ▷ Per SI S3
  until ComputeCV($\mathcal{M}$) < 0.01
  return $\mathcal{M}$
end procedure

S12. Theoretical Guarantees

S12.1 Expressivity and Approximation

Let p be a probability density on a compact set $\Omega \subset \mathbb{R}^d$ with a (piecewise) Lipschitz continuous log-density and bounded curvature of its principal level sets. For any $\varepsilon{>}0$, there exist finite

Algorithm 2 Rank-weighted neighbor updates

procedure ApplyRankWeightedUpdates($\mathbf{x}, \mathcal{M}, \mathcal{N}_k$)
  for $j = 1 \dots k$ **in** $\mathcal{N}_k$ (rank-ordered) **do**
    $w_j \leftarrow 2^{-(j-1)}$
    $\mathbf{node} \leftarrow \mathcal{N}_k[j]$
    $\mathbf{e} \leftarrow \mathbf{x} - \mathbf{node.w}$
    $\mathbf{node.m} \leftarrow (1 - \alpha w_j)\mathbf{node.m} + \alpha w_j\, \mathbf{e}$; $\mathbf{node.s} \leftarrow (1 - \alpha w_j)\mathbf{node.s} + \alpha w_j\, \mathbf{e}^{\circ 2}$
    $\rho \leftarrow \|\mathbf{node.m}\|/(\mathbf{node}.\sigma + \varepsilon)$; $\mathbf{node.a} \leftarrow \mathbf{node.a} + \eta_{cent}\, \rho\, \mathbf{node.m}$
    UpdateOja($\mathbf{node.u}, \mathbf{e}$)
    $\mathbf{node.h} \leftarrow \mathbf{node.h} + w_j$ ▷ Fractional hits and link counters
  end for
end procedure

Algorithm 3 Simplex-native updates (Stage 2)

procedure ApplySimplexNativeUpdates($\mathbf{x}, \mathcal{C}$)
  $C \leftarrow$ LocateWinningSimplex($\mathbf{x}$) ▷ Barycentric or nearest with projection
  Order vertices of C by increasing $\|\mathbf{x} - \mathbf{w}\_v\|$; assign rank-weights $w_j = 2^{-(j-1)}$
  for each vertex v of C in rank order **do**
    $\mathbf{e} \leftarrow \mathbf{x} - \mathbf{w}\_v$; $\mathbf{m}\_v \leftarrow (1 - \alpha w_j)\mathbf{m}\_v + \alpha w_j\, \mathbf{e}$; $\mathbf{s}\_v \leftarrow (1 - \alpha w_j)\mathbf{s}\_v + \alpha w_j\, \mathbf{e}^{\circ 2}$
    $\rho \leftarrow \|\mathbf{m}\_v\|/(\sigma\_v + \varepsilon)$; $\mathbf{a}\_v \leftarrow \mathbf{a}\_v + \eta\_cent\, \rho\, \mathbf{m}\_v$
  end for
  Accumulate facet pressures for C per S5
end procedure

budgets for recursion depth, mesh resolution, and non-linear warps (mini-NSFs) such that the model density q learned by the Proteus procedure satisfies $\|p - q\|_{L_1} < \varepsilon$.

The proof is constructive, following the algorithmic stages. We show that the total approximation error can be bounded by the sum of errors from a hierarchical decomposition, a piecewise-linear approximation of each component, and a non-linear residual correction. By making the resolution and capacity of each stage sufficient, the total error can be made arbitrarily small.

S12.2 Correspondence: Simplex Equilibrium Implies Effective Node Equilibrium

Setting and assumptions. Consider a simplicial complex $\mathcal{K} = (\mathcal{V}, \mathcal{S})$ with $|\mathcal{V}| = N$ nodes and $|\mathcal{S}| = M$ top-dimensional simplexes (each with $d{+}1$ vertices), nondegenerate simplex stars around each node, locally stationary sampling, and a fixed characteristic scale τ^*. Let P be the data-generating measure with $\mathrm{supp}(P) \subset \bigcup_{C \in \mathcal{S}} C$. For node i let V_i denote the Voronoi cell of w_i and write $P(V_i)$ for its mass; for $C \in \mathcal{S}$ write $P(C)$ for its mass. Empirical estimates from H i.i.d. samples are $\hat{P}(V_i) = h_i/H$ and $\hat{P}(C) = h_C/H$.

Claim. Driving the system toward simplex equilibrium ($\hat{P}(C) \approx 1/M$ for all simplexes) induces effective node equilibrium ($\hat{P}(V_i) \approx 1/N$) up to (i) an irreducible structural term controlled by simplex shape quality and star regularity, (ii) a residual equilibration tolerance, and (iii) a sampling term that vanishes as $H \to \infty$. The structural residual is exactly the dimensionality-junction signature.

Mechanism. The dominant equilibration channel is the *variance-cap split* $\sigma_i^2 > \tau_{\mathrm{local},i}$ (Algorithm 1), gated by the local evidence test of S3.4; the Torsion Ladder is a secondary refinement that redirects, rather than triggers, splits in curved regions. Decompose any candidate over-mass $P(V_i) \gg 1/N$ into two regimes:

- *Centering imbalance.* If $\mathbf{m}_i := \mathbb{E}[\mathbf{x} - \mathbf{w}_i \mid \mathbf{x} \in V_i] \neq 0$, the centering update $\eta_{\text{cent}}\rho_i\mathbf{m}_i$ in Algorithm 2 drives $\mathbf{m}_i \to 0$. This step is *approximately volume-preserving for* V_i (slow geometry, S2.3) and so cannot by itself reduce $P(V_i)$; it only ensures subsequent equilibration acts on cell volume rather than position bias.

- *Mass (cell-volume) imbalance.* With $\mathbf{m}_i \approx 0$, persistent over-mass implies V_i is geometrically over-extended at scale τ^*. Under the conditions of S14, $\sigma_i^2 \to \text{tr}(C_i)$, the trace of the Gaussian-windowed local covariance at $\mathbf{w}_i$ at bandwidth $\sigma = \sqrt{\tau}/c_{d,k}$. An over-extended cell elevates this trace through the kernel-weighted contribution of mass at distance $\gtrsim \sigma$ from $\mathbf{w}_i$; the variance cap fires, and a split is proposed along the dominant Oja axis (S8.1). The flow-conserving initialization (S11) transfers a strictly positive mass increment from V_i to the new child V_j.

The Torsion Ladder enters only when the over-mass coexists with non-PL-resolvable curvature: the proposed split is then redirected along $\text{ax}(\Omega_S)$ rather than the Oja axis (S4). The S3.4 evidence gate filters every proposed split, so the loop terminates when no further split improves local evidence (i.e., when the proposed children carry insufficient transition support to ratify the topology change).

Negative feedback (informal). Iterating the propose–gate–accept loop over all overloaded nodes drives $\sup_i P(V_i)$ monotonically downward in expectation: each accepted split removes one node from the overloaded set while injecting bounded amounts into neighboring cells, and the variance-cap rule guarantees that further splits are proposed at any node still violating $\sigma_i^2 \leq \tau_{\text{local},i}$. A formal Lyapunov function for the discrete-time, evidence-gated dynamics is not constructed here; see S15.1.

Why the loop stalls at junctions. At a k-D / ℓ-D dimensionality junction with $k < \ell$ meeting at node i, the local mass distribution is anisotropic and partially supported on a lower-dimensional subset. The variance trace $\text{tr}(C_i)$ saturates at a value reflecting the higher-dimensional limb's spread; additional splits proposed along the lower-D limb either fall below the cap (fewer effective directions of dispersion) or fail the S3.4 evidence test (the lower-D limb cannot generate enough transition counts to ratify added structure). The simplex star around i then carries a permanent dimensional mismatch: incident simplexes span different limb subsets, yielding non-uniform $|\mathcal{S}(i)|$ and bimodal $\alpha_{i,C}$. The structural floor of the Theorem above is precisely the residual node-mass imbalance this configuration produces — the loop's stalled state is the dimensionality-junction signature. A formal stuck-junction lemma is left open (S15.2).

Concentration form (static correspondence). The mechanism above describes *why* the algorithm drives simplex masses toward $1/M$; the following result quantifies *the consequence*: any state in which simplex masses are equilibrated to within tolerance ϵ_{sim} implies node masses equilibrated to within an explicit, finite-sample bound. The static correspondence requires only the Voronoi–simplex mass identity and multinomial concentration; it does not depend on the specifics of the equilibration dynamics.

Lemma 1 (Voronoi–simplex decomposition; deterministic). Define the barycentric Voronoi share of vertex i within simplex $C \ni i$ by

$$\alpha_{i,C} \;:=\; P(V_i \cap C)/P(C)\,, \qquad \sum_{i \in V(C)} \alpha_{i,C} \;=\; 1\,.$$

Since $\{V_i\}$ partitions $\mathrm{supp}(P)$ and $\{C\}$ tiles it,

$$P(V_i) \; = \; \sum_{C \ni i} \alpha_{i,C}\, P(C)\,. \tag{D}$$

Lemma 2 (shape and star regularity). Under simplex shape quality $Q_S \geq Q_{\min}$ enforced by the Torsion Ladder (S4) and pruning gauntlets (S3), there exists $\beta_{\max} = \beta_{\max}(d, Q_{\min})$ with $\beta_{\max}(d, 1) = 0$ such that

$$\left|\alpha_{i,C} - \tfrac{1}{d+1}\right| \;\leq\; \beta_{\max} \quad \text{for every incident pair } (i, C)\,.$$

Counting incidences gives $\sum_i s_i = (d{+}1)M$, so $\bar{s} := (d{+}1)M/N$. Assume star regularity $|s_i/\bar{s} - 1| \leq \eta_s$ for all i (a property the algorithm maintains via simplex-arbitrated pruning, S3.3).

Lemma 3 (multinomial concentration). Let h_C be multinomial counts over $\mathcal{S}$ from H i.i.d. draws. By Hoeffding with a union bound, with probability $\geq 1 - \delta$,

$$\max_{C \in \mathcal{S}} \left|\hat{P}(C) - P(C)\right| \;\leq\; \Delta_M(H, \delta) \;:=\; \sqrt{\tfrac{\log(2M/\delta)}{2H}}\,. \tag{$\mathrm{C_S}$}$$

The same bound applies to the N-cell Voronoi multinomial,

$$\max_{i \in \mathcal{V}} \left|\hat{P}(V_i) - P(V_i)\right| \;\leq\; \Delta_N(H, \delta) \;:=\; \sqrt{\tfrac{\log(2N/\delta)}{2H}}\,. \tag{$\mathrm{C_V}$}$$

Bernstein refinements give the tighter $\sqrt{2\log(2M/\delta)/(MH)} + \log(2M/\delta)/(3H)$ when $P(C) \approx 1/M$ (and analogously for the Voronoi cells); we use the Hoeffding form below for clarity.

Theorem (effective node equilibrium, finite-sample). Suppose the algorithm achieves simplex equilibrium with empirical tolerance

$$\left|\hat{P}(C) - 1/M\right| \;\leq\; \epsilon_{\mathrm{sim}}/M \quad \text{for all } C \in \mathcal{S}\,,$$

and that Lemmas 1–2 hold. Then on the joint event of (**??**)–(**??**) (probability $\geq 1 - 2\delta$), for every node i,

$$\left|\hat{P}(V_i) - \tfrac{1}{N}\right| \;\leq\; \underbrace{\frac{\eta_s + (1{+}\eta_s)(d{+}1)\beta_{\max}}{N}}_{\text{structural}} \;+\; \underbrace{\overline{\alpha}_d\, s_i\, \frac{\epsilon_{\mathrm{sim}}}{M}}_{\text{equilibrium}} \;+\; \underbrace{\overline{\alpha}_d\, s_i\, \Delta_M(H,\delta) + \Delta_N(H,\delta)}_{\text{sampling}}, \tag{T}$$

where $\overline{\alpha}_d = 1/(d{+}1) + \beta_{\max}$.

Proof. Decompose the deviation as

$$\hat{P}(V_i) - \tfrac{1}{N} \;=\; \left[\hat{P}(V_i) - P(V_i)\right] \;+\; \left[P(V_i) - \tfrac{1}{N}\right].$$

The first bracket is bounded by Δ_N via (**??**). For the second, apply (**??**) and write $\alpha_{i,C} = \frac{1}{d+1} + \beta_{i,C}$ with $|\beta_{i,C}| \leq \beta_{\max}$, and $s_i = (1{+}\xi_i)\bar{s}$ with $|\xi_i| \leq \eta_s$:

$$P(V_i) - \tfrac{1}{N} \;=\; \left[\Big(\tfrac{s_i}{d+1} + \sum_{C \ni i} \beta_{i,C}\Big)\tfrac{1}{M} - \tfrac{1}{N}\right] + \sum_{C \ni i} \alpha_{i,C}\big(P(C) - \tfrac{1}{M}\big)\,.$$

Using $\bar{s}/M = (d{+}1)/N$, the first bracket equals $\xi_i/N + (1/M)\sum_{C \ni i} \beta_{i,C}$, whose absolute value is at most $\eta_s/N + s_i\beta_{\max}/M \leq \eta_s/N + (1{+}\eta_s)(d{+}1)\beta_{\max}/N$ — the structural term. For the second sum, combine the algorithmic tolerance with (**??**):

$$\left|P(C) - 1/M\right| \;\leq\; \left|\hat{P}(C) - 1/M\right| + \left|\hat{P}(C) - P(C)\right| \;\leq\; \epsilon_{\mathrm{sim}}/M + \Delta_M\,,$$

and $\sum_{C \ni i} \alpha_{i,C} \leq \overline{\alpha}_d s_i$. Combining yields (**??**). $\square$

Interpretation. The bound separates three sources of node-mass deviation, each with its own scaling:

1. *Structural* $(\eta_s + (d{+}1)\beta_{\max})/N$: irreducible without further mesh edits, controlled by simplex shape quality and star regularity. Both quantities are actively bounded by the Torsion Ladder (S4), shape-quality safeguards ($Q_S \geq 0.25$), and simplex-arbitrated pruning (S3.3). In the regular limit ($Q_S \to 1$, $\eta_s \to 0$) this term vanishes.

2. *Equilibrium* $\overline{\alpha}_d s_i\, \epsilon_{\text{sim}}/M \approx (1{+}\eta_s)\epsilon_{\text{sim}}/N$: the residual algorithmic equilibration tolerance, which decreases as splits/merges drive empirical simplex masses toward $1/M$. The S3.4 evidence gating ensures monotone improvement in ϵ_{sim} on accepted edits.

3. *Sampling* $\overline{\alpha}_d s_i\, \Delta_M {+} \Delta_N = \mathcal{O}\big(M\sqrt{\log(M/\delta)/H}/N {+} \sqrt{\log(N/\delta)/H}\big)$: vanishes as $H \to \infty$. The $M\Delta_M/N$ scaling tightens to $\sqrt{M\log(M/\delta)/(N^2H)}$ under the Bernstein refinement; under either scaling, $H \gg M$ samples suffice for the term to drop below the structural floor.

In the asymptotic regime ($H \to \infty$, $\epsilon_{\text{sim}} \to 0$) the bound reduces to the irreducible structural term. Hence *any* persistent deviation $|\hat{P}(V_i) - 1/N| = \Omega(1/N)$ that survives both the sampling and equilibration limits must be supported by elevated η_s or $\beta_{\max}$ in the local star — i.e. by genuine geometric irregularity. This is the negative-feedback content of the sketch made quantitative: the algorithm's only mechanism for sustaining a node-mass imbalance against its torsion/evidence controllers is to maintain a structurally irregular star, which the data themselves enforce only at intrinsic-dimensional junctions or other topological features.

Remarks. The argument uses only local stationarity, nondegenerate stars, and standard multinomial concentration. The structural floor in (**??**) furnishes an *upper bound on residual imbalance after equilibration*; conversely, observing $|\hat{P}(V_i) - 1/N|$ above the sampling and equilibrium scales is a *lower bound* on local irregularity (η_s or $\beta_{\max}$), which is the formal content of the dimensionality-junction signature. Empirically, the correspondence manifests around overloaded nodes as: sharply bimodal simplex activity within the node's star at dimensionality junctions; a clean partition of link statistics into high-significance backbone links versus low-significance bridge links; and bridge links that exhibit persistent directional hit asymmetry together with low net face pressure, distinguishing stable junctions from active frontiers.

The Signature of a Dimensionality Junction. The "informative mismatch" at structural junctions manifests as a concrete, stable, and observable statistical signature. A node located where, for example, a 1D filament meets a 2D sheet will persistently exhibit:

- **Bimodal Simplex Activity:** The distribution of internal activity values for its adjacent simplexes will be sharply bimodal.
- **Divergent Link Significance:** Its links will partition into high-significance "backbone" links (within a manifold) and low-significance "bridge" links (connecting manifolds).
- **Stable Asymmetry on Bridge Links:** Bridge links show high, stable directional hit-counter asymmetry but, crucially, low net face pressure. This distinguishes a stable junction from an active, growing frontier, where both asymmetry and pressure would be high.

This emergent signature is not a failure of the algorithm, but a feature that allows the system to report on the complex local topology of the data.

S13. The Proteus Generative Model and Free-Energy Objective

This section states the statistical model that underlies Proteus and provides the Bayesian model-selection objective used to justify discrete structure changes under finite data. The key idea is that the learned topology induces a geometry-derived router $q(j \mid i; m)$, and the observed transition counts $\{n_{i\to j}\}$ act as evidence about a latent simplex mass field m. Continuous updates (settling) reduce transition prediction error, while discrete edits (splits/merges/pruning) are accepted only when they improve the local evidence score of S3.4 (a Laplace-approximated log marginal likelihood; "free-energy proxy" in the broader literature, but referring here to Bayesian evidence, not to variational free energy).

S13.1 Objects and Geometry-Induced Conditionals

Let $X = \{x_n\}_{n=1}^N \subset \mathbb{R}^d$ and let $G = (\mathcal{V}, \mathcal{E})$ denote the Hebbian/GNG graph with node locations $\{z_i\}$. Per-node BMU hit counts are n_i and directional edge counts are $n_{i\to j}$ (first/second NN "conditional hits"). Define empirical conditionals $\hat{p}(j \mid i) = n_{i\to j}/\max(1, n_i)$.

Let $\mathcal{S}$ denote the dual (Delaunay-like) complex with simplexes $S \in \mathcal{S}$ that tile the space between adjacent nodes. We introduce unknown simplex masses $m = \{m_S\}_{S\in\mathcal{S}}$ with $m_S \geq 0$ and $\sum_S m_S = 1$. Geometry induces barycentric weights $w_{iS} \in [0, 1]$ with $\sum_{S\ni i} w_{iS} = 1$, and the mass "seen" by node i is

$$M_i(m) = \sum_{S\ni i} w_{iS}\, m_S\,.$$

A geometry-derived conditional from $i \to j$ implied by m is

$$q(j \mid i; m) = \frac{\sum_{S\ni i,j}\ \kappa_{ijS}\, m_S}{\sum_{S\ni i}\ \kappa_{iS}\, m_S}\,.$$

Canonical choice for κ (committed). For every d-simplex S with vertex $i \in S$, let A_{iS} denote the $(d-1)$-volume of the facet of S *opposite* to i (equivalently, $A_{iS} = d \cdot |S|_d/h_{iS}$ where $|S|_d$ is the d-volume of S and h_{iS} is the altitude from i). Define

$$\kappa_{iS} \;=\; A_{iS}\,, \qquad \kappa_{ijS} \;=\; \frac{A_{iS}}{d} \quad \text{for } j \in S,\ j \neq i\,.$$

This is the "uniform facet-share" choice: the total conductance κ_{iS} of node i into simplex S equals the opposite-facet area, and it is split uniformly among the d neighbors of i within S. Consistency: $\sum_{j\in S,\, j\neq i} \kappa_{ijS} = d \cdot A_{iS}/d = \kappa_{iS}$, so the router normalizer is always the sum of conductances leaving i. The choice is manifestly positive on nondegenerate complexes, scale-covariant in the mesh geometry, and invariant under simplex relabeling/orientation. Alternative weightings (cotangent/FEM stiffness, altitude-ratio, edge-length-scaled facet area) preserve the identifiability analysis below; we commit to uniform facet-share as the default because it is cheap to compute, avoids sign issues on obtuse simplices, and yields a clean star-matrix structure (S13.7).

S13.2 Likelihood from Dirichlet–Multinomial Evidence

Treat outgoing counts at node i as multinomial draws given $q(\cdot \mid i; m)$:

$$\{n_{i\to j}\}_j \,\big|\, n_i,\, m \sim \text{Multinomial}\big(n_i,\, q(\cdot \mid i; m)\big)\,.$$

Placing a small Dirichlet smoothing prior on $q(\cdot \mid i)$ and integrating it out yields a Dirichlet–multinomial factor at node i,

$$\phi_i(m) \;\propto\; \prod_{j\in\mathcal{N}(i)} q(j \mid i; m)^{\,n_{i\to j}}\,,$$

up to a gamma-ratio constant independent of m. Thus each node contributes a log-linear potential in $\log q(j \mid i; m)$: Hebbian counts provide conjugate evidence about local transitions, while m is the global mass field that must rationalize all local transitions via geometry.

S13.3 Priors on the Mass Field m

Several Bayesian priors are natural, each imparting a different character to the model:

- **Simplex-wide Dirichlet (composition).** $m \sim \mathrm{Dirichlet}(\beta_S)_{S \in \mathcal{S}}$. Captures uncertainty on a fixed partition. It is conjugate if counts were observed directly in cells; here, the link is indirect via the transition likelihood, but the model remains coherent.
- **Logistic-Gaussian MRF (smoothness).** Let $\theta_S \in \mathbb{R}$ with GMRF prior $\theta \sim \mathcal{N}(0,\, \tau^{-1} L^{+})$ on the dual-graph Laplacian L, and set $m_S = \exp(\theta_S)/\sum_{S'} \exp(\theta_{S'})$. This is a logistic-normal prior that encourages neighboring simplexes to have similar mass, inducing spatial smoothness.
- **Dirichlet/Pólya tree on a mesh (hierarchical composition).** Impose a hierarchical split scheme over $\mathcal{S}$ (e.g., via a spanning tree) and place Dirichlet priors at each split. This defines a non-parametric model over recursive partitions, where the partition itself is learned by the Hebbian process.
- **Normalized random measures.** Assign independent $\tilde{m}_S \sim \mathrm{Gamma}(a_S, 1)$ and set $m_S = \tilde{m}_S / \sum_{S'} \tilde{m}_{S'}$. This is equivalent to a Dirichlet with parameters a_S on a fixed set $\mathcal{S}$ and connects the model to the broader theory of completely random measures.

S13.4 Posterior as a Factor Graph and Sum–Product

The posterior over m (up to normalization) is

$$p(m \mid \mathrm{data}) \;\propto\; \underbrace{\prod_{i \in \mathcal{V}} \phi_i(m)}_{\text{Dirichlet–multinomial edge evidence}} \;\times\; \underbrace{\psi(m)}_{\text{prior on } m},$$

where ψ is one of the priors above. Sum–product (belief propagation) on the dual factor graph performs approximate marginalization of this posterior. With a logistic-GMRF prior, this becomes a CRF-like model with spatial smoothness; with a Dirichlet prior, it is a compositional model without coupling.

S13.5 MaxEnt/MAP Connection

Setting a flat prior on m and maximizing the log-likelihood yields the MaxEnt/MLE estimator

$$\hat{m} \;=\; \arg\max_{m \in \Delta} \sum_i \sum_j n_{i \to j} \, \log q(j \mid i; m)\,.$$

Adding $\log \psi(m)$ yields a MAP estimator under the chosen Bayesian prior. Thus the Proteus MaxEnt + sum–product pipeline is the zero-prior (or uniform-prior) MAP limit of this coherent Bayesian model.

S13.6 Relation to BNP Families and Multiscale Refinement

The Proteus framework synthesizes ideas from several BNP families, but its data-driven, geometric construction creates key differences.

At a fixed scale, Proteus acts as a partition-based BNP model on a learned simplicial mesh. Its closest cousins are:

- **Pólya/Optional Pólya Trees (PT/OPT):** These models define densities on recursive, axis-aligned partitions. Proteus is similar in using a partition, but its partition is a *data-learned simplicial mesh with cycles*, not an axis-aligned tree. Furthermore, its likelihood is based on *transition counts* between nodes, not direct bin counts.
- **Log-Gaussian Cox Processes (LGCP):** These models use a latent Gaussian field to define a smooth log-density. Proteus with a logistic-GMRF prior is similar, but its likelihood is a CRF-like potential based on transition counts, coupling neighboring cells via $q(j \mid i; m)$, whereas LGCPs typically use direct Poisson counts in fixed bins.
- **Dependent Dirichlet Processes (DDP):** These models allow mass allocation to vary over space. However, DDPs typically randomize mixture weights and atoms in an infinite-dimensional space, whereas Proteus defines a *finite-dimensional* mass vector m whose components are geometrically coupled by the learned Hebbian topology.

Across scales, the coarse→fine recursion resembles a hierarchical BNP prior. For a coarse cell $P \in \mathcal{S}^{(s)}$ with children $\mathrm{ch}(P) \subset \mathcal{S}^{(s+1)}$, the refinement can be modeled as a stochastic process:

$$m^{(s+1)}_{\mathrm{ch}(P)} \mid m^{(s)}_P \;\sim\; \mathrm{Dir}\big(\alpha_s\, \pi^{(s)}_{\mathrm{ch}(P)}\, m^{(s)}_P\big)\,, \qquad \sum_{C \in \mathrm{ch}(P)} m^{(s+1)}_C = m^{(s)}_P\,.$$

This is the multiscale Dirichlet refinement used by Pólya trees, but applied to a simplicial mesh. The warm-start procedure in Proteus, where the converged state at scale s initializes the search at $s+1$, is a deterministic analogue of using the posterior at one level as the prior for the next. The variance-based splitting rule acts as a data-driven control on mesh refinement, analogous to a Bayes factor test for splitting a cell.

S13.7 Star Matrix and Identifiability Theorem

Under the canonical κ of S13.1, the router at node i has a clean linear structure. Let $\mathrm{Star}(i) = \{S \in \mathcal{S} : S \ni i\}$ and $\mathcal{N}(i) = \{j \neq i : \exists S \in \mathrm{Star}(i),\, j \in S\}$. Stack the masses in $\mathrm{Star}(i)$ as $m^{(i)} \in \mathbb{R}^{|\mathrm{Star}(i)|}_{\geq 0}$, ordered by some fixed indexing of $\mathrm{Star}(i)$.

Star matrix. Define $K_i \in \mathbb{R}^{|\mathcal{N}(i)| \times |\mathrm{Star}(i)|}$ by

$$(K_i)_{jS} \;=\; \begin{cases} A_{iS}/d & \text{if } j \in S\,, \\ 0 & \text{otherwise,} \end{cases} \qquad k_i \;=\; K_i^\top \mathbf{1}_{|\mathcal{N}(i)|}\,,$$

where $\mathbf{1}$ is the all-ones vector. By construction $(k_i)_S = \sum_{j \in \mathcal{N}(i)} (K_i)_{jS} = d \cdot A_{iS}/d = A_{iS} = \kappa_{iS}$, confirming the row sums. The router at i reads

$$q(\cdot \mid i; m) \;=\; \frac{K_i\, m^{(i)}}{k_i^\top m^{(i)}} \;\in\; \Delta^{|\mathcal{N}(i)|-1}\,.$$

Theorem (local and global identifiability). Assume

[leftmargin=1.5em]

(A1) *Local rank condition:* for every i, $\ker(K_i) \cap \mathrm{span}(k_i)^\perp = \{0\}$, i.e. K_i has rank $|\mathrm{Star}(i)|$ on the complement of the one-dimensional scaling subspace.

(A2) *Star overlap / connected dual:* the dual graph on $\mathcal{S}$ (simplices adjacent by shared facet) is connected.

Then m is uniquely determined by the full set of transition probabilities $\{q(\cdot \mid i; m)\}_{i \in \mathcal{V}}$ up to the simplex constraint $\sum_S m_S = 1$.

Proof sketch. Fix i. If $q(\cdot \mid i; m) = q(\cdot \mid i; m')$ then $K_i m^{(i)}$ and $K_i m'^{(i)}$ are positively proportional: $K_i(m^{(i)} - \lambda_i m'^{(i)}) = 0$ with $\lambda_i = (k_i^\top m^{(i)})/(k_i^\top m'^{(i)})$. Assumption (A1) forces $m^{(i)} = \lambda_i m'^{(i)}$ on $\mathrm{Star}(i)$. For two stars sharing a simplex S, the two scalars λ_i, λ_j must agree on m_S, giving $\lambda_i = \lambda_j$. By (A2), all scalars agree: $m = \lambda m'$ globally. The simplex constraint then forces $\lambda = 1$. $\square$

When does (A1) hold? A counting bound. K_i has $|\mathrm{Star}(i)|$ columns and $|\mathcal{N}(i)|$ rows. Each column (simplex S) has exactly d non-zeros (its d other vertices). A necessary condition for full column rank is $|\mathcal{N}(i)| \geq |\mathrm{Star}(i)|$. For generic manifolds at their intrinsic dimension, a node with k neighbors of degree d typically has $|\mathrm{Star}(i)| \approx k$ and $|\mathcal{N}(i)| \approx k$; the condition is tight but usually met. Two regimes where it fails:

[leftmargin=1.5em]

- **Degenerate simplices.** $A_{iS} \to 0$ (sliver simplex, altitude collapse) zeroes an entire column of K_i, making that column linearly dependent on zero. Detected by the shape-quality guard $Q_S \geq Q_{\min}$ (S4).
- **Collinear neighbor rays.** If two simplices S_1, S_2 of $\mathrm{Star}(i)$ have identical vertex sets apart from i itself (possible only on thin/non-manifold configurations), their columns of K_i are proportional. Detected by simplex-enumeration deduplication.

Neither regime arises in a well-formed simplicial complex with bounded shape quality. We state (A1) as a runtime hypothesis rather than a universal guarantee.

Runtime conditioning check. Compute $\sigma_{\min}(K_i)/\sigma_{\max}(K_i)$ at each node whose star has just changed. The check is cheap ($\mathcal{O}(|\mathrm{Star}(i)|\,|\mathcal{N}(i)|^2)$ via thin SVD, amortized over simplex edits). If the ratio falls below a threshold $\rho_{\min}$ (default 10^{-4}; conservative 10^{-3}):

[leftmargin=1.5em]

1. Flag i as *router-ill-conditioned*.
2. Skip the MAP settle contribution from node i in the local evidence score $F(\mathcal{R}; M)$ of S3.4 (equivalently, set $n_{i\to j}$ to zero at i for that scoring pass).
3. Fall back to purely geometric acceptance for any edit whose affected region $\mathcal{R}$ contains i, until the condition is restored (typically on the next local edit that resolves the degeneracy).

This fallback is strictly more conservative than the evidence gate: it never *accepts* an edit it would otherwise reject; it only declines to use the likelihood contribution from an ill-conditioned neighborhood.

Connection to Fisher information. K_i is (up to normalization) the Jacobian of $q(\cdot \mid i; m)$ with respect to $m^{(i)}$ at the canonical κ. The local Fisher information matrix of the transition likelihood $\phi_i(m)$ is $I_i(m) = n_i\, K_i^\top \mathrm{diag}(1/q)\, K_i$ (up to the normalizer). (A1) is precisely the condition that $I_i(m)$ is nonsingular on the tangent of the simplex at m. This gives a cleanly asymptotic version of identifiability: under (A1–A2), the Fisher information of the full transition model is nonsingular on $\{\sum_S m_S = 1\}$, and standard MLE consistency applies as $\min_i n_i \to \infty$.

S13.8 Practical Notes

Consistency and identifiability. Under canonical κ (S13.1) and the star-rank condition (A1), the posterior over m concentrates as $n_{i\to j}$ grows. A weak smoothness prior (e.g., logistic-GMRF) regularizes marginal cases where (A1) is borderline. **Uncertainty propagation.** A fully Bayesian treatment propagates finite-sample noise in $n_{i\to j}$ into credible intervals over m; the MAP approach of S3.4 is the zero-prior limit. **Diagnostic logging.** The runtime conditioning check (S13.7) doubles as a numerical health diagnostic: persistent ill-conditioning at a vertex i is a signal that the local complex has degenerated (slivers, accidental collinearities, broken stars) and should be flagged for Stage 2 shape-quality remediation.

S13.9 Summary

The Proteus pipeline corresponds to the following Bayesian model and inference loop:

- **Hebbian Learning/GNG** $\to$ A method for learning a **data-driven prior structure** (the simplicial partition $\mathcal{S}$).
- **Edge Hit Statistics ($n_{i\to j}$)** $\to$ Evidence for **Dirichlet–multinomial local likelihoods** ($\phi_i(m)$).
- **Dual-Graph Sum–Product** $\to$ An algorithm for **approximate Bayesian marginalization** over the global simplex mass field m.
- **MaxEnt Objective** $\to$ The **MAP inference** limit of the same model under a uniform prior.

Proteus uses geometric signals to propose discrete edits to $\mathcal{S}$, and the local evidence score of S3.4 to accept edits only when they improve the approximate log marginal likelihood after accounting for model complexity.

S14. Conditions for Equivalence of Stage 1 to Gaussian-Weighted PCA/SVD

Claim (informal). Under standard limits—frozen topology, infinitesimal step-size, stable neighborhoods, and infinite data—the Stage 1 moment tracker with rank-weighted updates and Oja's rule performs *local* Gaussian-weighted PCA/SVD at node locations. The variance threshold acts as a scale-dependent cap that adaptively refines regions whose Gaussian-windowed variance exceeds the target.

S14.1 Setting and Kernel Mapping

Let K_σ denote an isotropic Gaussian window with bandwidth σ, and let w_i be a (temporarily fixed) node location. Using the scale calibration in S2.1, set

$$\sigma \;=\; \sqrt{\tau}/c_{d,k}\,,$$

so that the per-node statistics at w_i target the Gaussian-smoothed local moments at the Stage 1 threshold τ.

Define the Gaussian-windowed local mean and covariance centered at w_i:

$$\mu_i \;=\; \mathbb{E}_{K_\sigma(\cdot - w_i)}[x]\,, \qquad C_i \;=\; \mathbb{E}_{K_\sigma(\cdot - w_i)}\big[(x-\mu_i)(x-\mu_i)^\top\big].$$

S14.2 Conditions for Equivalence

The following constraints yield the reduction of Stage 1 statistics to Gaussian-weighted PCA/SVD:

1. **Frozen topology and slow geometry.** During the estimation window, disable splits/pruning and take "slow" geometric motion (deferred moves with $\eta_{\text{cent}} \to 0$) so positions change on a slower time scale than moment convergence (cf. S2.3 deferred nudge).
2. **Infinitesimal step-size, infinite data.** Use $\alpha \to 0$ with a Robbins–Monro schedule or a small constant EWMA weight (S2.2) and an effectively infinite stream of i.i.d. samples from $p(x)$. Then EWMAs converge to the corresponding kernel expectations.
3. **Stable, approximately isotropic neighborhoods.** Use ANN settings where k is large enough and efSearch sufficiently high so the rank-ordered neighbor sets approximate samples from an isotropic ball around w_i of radius $\mathcal{O}(\sigma)$. The rank-geometric weights $2^{-(j-1)}$ then act as a discrete radial kernel that weakly converges to K_σ as $N \to \infty$, $k \to \infty$, $k/N \to 0$.
4. **Oja's rule for the principal subspace.** Apply standard Oja updates to residuals $e = x - w_i$ (cf. S8.1). Under the above sampling and step-size limits, Oja converges almost surely to the principal subspace of C_i; the tracked eigenpairs match those from the SVD of the local Gaussian-weighted covariance.
5. **Variance-cap thresholding.** Use the split criterion $\sigma_i^2 > \tau_{\text{local},i}$ with $\tau_{\text{local},i}$ mapped from the control per S2.4. With EWMAs equal to kernel moments, this is equivalent to adaptively refining where $\text{tr}(C_i)$ exceeds the target scale.

S14.3 Consequences

Under these conditions, the Stage 1 per-node estimates equal the Gaussian-windowed moments at w_i:

$$m_i \to \mu_i - w_i\,, \qquad \sigma_i^2 \to \text{tr}(C_i)\,; \quad \text{Oja}(e) \;\Rightarrow\; \text{eigvectors/values of } C_i.$$

Thus, the procedure performs *local PCA/SVD at scale σ*, and the thresholded growth rule implements an adaptive partition until the local Gaussian-smoothed variance falls below the cap.

S14.4 Explicit Limits Recovering SVD

- **Global PCA limit.** Single node, $k \to N$, and σ larger than the data diameter $\Rightarrow$ uniform weights over the dataset; the covariance reduces to the global covariance and Oja recovers global PCA.
- **Local PCA limit.** Fixed nodes, $\alpha \to 0$, large k with isotropic neighborhoods $\Rightarrow$ each node estimates the SVD of C_i, the Gaussian-windowed local covariance at w_i.
- **Piecewise-linear manifold limit.** With growth enabled at fixed τ, the mesh partitions the space so each cell maintains $\text{tr}(C_i) \le \tau_{\text{local},i}$; within cells, local SVD is recovered and provides the dominant direction for geometric refinement (via Oja).

S14.5 What Is *Not* the Same

- With active growth/topology changes, the method is *not* a single global SVD of the convolved dataset; it is a stitched collection of local Gaussian-weighted SVDs with adaptive centers and supports.

- At finite k, rank-geometric weights only approximate Gaussian radial weighting; exact equality holds asymptotically under the neighborhood stability conditions.
- Stage 2 constructs (simplicial complex, torsion audit, dual flow) have no SVD reduction; the equivalence pertains solely to Stage 1 at fixed scale.

S14.6 Takeaway

With the calibration $\sigma = \sqrt{\tau}/c_{d,k}$ (S2.1) and the constraints above, Stage 1 reduces to **local Gaussian-weighted PCA/SVD**, while the variance threshold implements **scale-aware adaptive refinement**. This provides a precise bridge between the Hebbian/GNG dynamics and kernel PCA at a chosen characteristic scale.

S15. Open Theoretical Questions

The arguments in S12 and S14 are intentionally a mixture of rigorous and informal. This section catalogues the remaining theoretical gaps so that future work can target them directly. Each item identifies (a) the claim, (b) the current status, and (c) what would be required to close it.

S15.1 Lyapunov function for the variance-cap / evidence-gated dynamics

Claim (S12.2 informal). Iterating the variance-cap split rule under S3.4 evidence gating drives $\sup_i P(V_i) \to 1/N$ up to the structural floor of the Theorem in S12.2.

Status. Per-step monotonicity is asserted by inspection: an accepted split at node i removes one violator from the overloaded set $\{i : \sigma_i^2 > \tau_{\text{local},i}\}$ and transfers a strictly positive mass increment to a child cell. However, three obstacles prevent a clean Lyapunov argument:

- *EWMA noise.* Variance estimates σ_i^2 are noisy EWMAs (S2.2); a single split may transiently raise neighboring σ_j^2 via the flow-conserving re-initialization (S11) before the EWMA decays.
- *Evidence rejections.* Splits failing the S3.4 test do not reduce the overloaded set, so the dynamics are not a strict descent on the cardinality of violators; they are a descent on the subset *accepted* splits can reduce.
- *Boundary effects between scales.* Across recursion levels the threshold $\tau_{\text{local},i}$ changes (S2.4), which can re-introduce violators that were previously satisfied.

What would close it. Two candidates:

1. A potential function of the form $\Phi = \sum_i (\sigma_i^2 - \tau_{\text{local},i})_+$ with bounded EWMA-noise term, plus a martingale argument showing $\mathbb{E}[\Phi_{t+1} \mid \mathcal{F}_t] \leq \Phi_t - c \cdot \mathbf{1}[\text{accepted split}]$ for $c > 0$. The challenging step is bounding the EWMA-induced fluctuation independently of the local sample rate.
2. A stochastic-approximation argument under the conditions of S14 (slow geometry, $\alpha \to 0$, large k) that turns the EWMA into a deterministic ODE. In that limit a Lyapunov function on the continuous-time variance trace is straightforward; the open question is the strength of the discretization gap.

S15.2 Stuck-junction lemma

Claim (S12.2 informal). At a k-D / ℓ-D dimensionality junction ($k < \ell$), the equilibration loop stalls at a configuration whose star irregularity (η_s) and shape-quality deviation ($\beta_{\max}$) are bounded below by quantities depending on the local intrinsic-dimension contrast.

Status. The qualitative picture (variance saturation on the higher limb, evidence-test failure on the lower limb) is consistent with the algorithm's mechanics, but no formal lemma is proven. In particular:

- The claim that S3.4 *must* reject splits across the junction depends on the local transition-count distribution near the junction, which has not been characterized analytically.
- The structural floor in the S12.2 Theorem is an upper bound; converting it into a *lower* bound at junctions requires showing that the data themselves enforce $\eta_s, \beta_{\max} \geq c(k, \ell, d) > 0$ at any star spanning the dimensional contrast.

What would close it. A model-driven calculation of the local transition-count distribution at a piecewise-linear junction (e.g. a k-flat meeting an ℓ-flat in $\mathbb{R}^d$ at zero), evaluated against the S3.4 acceptance criterion. This should yield (i) an explicit lower bound on η_s and $\beta_{\max}$ as a function of the dimensional contrast $\ell - k$ and the local mass ratio, and (ii) a critical sample size below which evidence rejection holds with high probability. The form is similar to local hypothesis-testing power calculations and is tractable for canonical junction geometries.

S15.3 Approximation budgets in S12.1

Claim (S12.1). For any target L_1 error ε, finite budgets for recursion depth, mesh resolution, and mini-NSF capacity exist that achieve $\|p - q\|_{L_1} < \varepsilon$.

Status. The claim is constructive in spirit but the explicit budget functions (depth, N_{nodes}, NSF capacity) $\to$ ε are not derived. The argument decomposes the total error into hierarchical, PL, and non-linear residual components without quantifying the constants.

What would close it. Standard manifold-approximation results give the PL term explicitly: for a C^2 manifold of intrinsic dimension d_{intr}, N nodes uniformly distributed yield Hausdorff error $\mathcal{O}(N^{-1/d_{\text{intr}}})$ and mass error of the same order under the variance cap. The hierarchical and NSF residual terms are model-class dependent; for the specific NSF parameterization in S7, density-approximation bounds for spline flows would supply explicit constants. Wiring these together into a single quantitative theorem is an exercise in bookkeeping but has not been performed in this paper.

S15.4 Convergence of S3.4 evidence acceptance under finite data

Claim (S3.4, implicit). The local evidence test asymptotically accepts only structure changes that improve the model under the true data-generating distribution.

Status. S3.4 is presented as a per-decision criterion; its asymptotic correctness is plausible by analogy with BIC consistency results, but no formal statement is made. The Dirichlet–multinomial marginal-likelihood ratio (an alternative to BIC, noted in OPEN_ISSUES item 5) would be conjugate to the local likelihood and would inherit a cleaner consistency argument.

What would close it. A consistency theorem of the form: under regularity conditions on the geometry-induced router q, the S3.4 acceptance test selects the correct topology among any finite candidate set with probability tending to 1 as $H \to \infty$. The argument follows the standard BIC/MDL consistency template (Kass–Raftery) once the regularity of q is established.

S15.5 Identifiability of the simplex mass field m — RESOLVED (S13.7)

Status. Addressed by the canonical κ committed in S13.1 (uniform facet-share) and the Star Matrix Identifiability Theorem in S13.7. Under assumptions (A1) local star-rank and (A2) connected dual graph, m is identifiable from $\{q(\cdot \mid i; m)\}_i$ up to the simplex constraint. A runtime conditioning check based on $\sigma_{\min}(K_i)/\sigma_{\max}(K_i)$ detects violations of (A1) in practice (slivers, degenerate stars), with a conservative fallback that skips the likelihood contribution from ill-conditioned neighborhoods. The identifiability statement corresponds to nonsingularity of the Fisher information on the probability simplex.

Residual open point. The theorem is *local-to-global* but static: it gives uniqueness of m at a fixed topology. A dynamic version (identifiability preserved along trajectories of the variance-cap + evidence loop) would additionally require that edits do not transit through ill-conditioned configurations. In practice, the conditioning check + shape-quality guard jointly suffice, but a formal trajectory-level guarantee is not given.

S15.6 Summary

The paper's theoretical guarantees fall into three tiers:

- **Rigorous:** S12.2 Theorem (effective node equilibrium, finite-sample), S14 conditions for kernel-PCA equivalence in the appropriate limit, S13.7 Star Matrix Identifiability Theorem (static identifiability under canonical κ).
- **Constructive but quantitatively incomplete:** S12.1 expressivity (S15.3), S13's BNP correspondence.
- **Mechanism-level / informal:** S12.2 dynamic equilibration (S15.1, S15.2), S3.4 consistency (S15.4), dynamic preservation of identifiability along trajectories (residual in S15.5).

The most actionable remaining items for future theory work are S15.1 (Lyapunov) and S15.4 (S3.4 consistency); both follow established templates and would together upgrade the algorithm's correctness story from "mechanism plus static correspondence plus static identifiability" to "mechanism plus dynamic convergence with preserved identifiability."